Types of Neural Networks *A theoretical overview of the major architectures*

Feedforward Neural Network (FNN) — `the baseline` The simplest form of a neural network. Data flows in one direction — from input to hidden layers to output — with no loops or cycles. Each neuron in one layer connects to every neuron in the next, which is why these are also called fully connected or dense networks. They are universal function approximators, meaning given enough neurons they can model any relationship, but they scale poorly with high-dimensional data like images or sequences.

Convolutional Neural Network (CNN) — `spatial data` Designed specifically for grid-like data such as images. Instead of connecting every neuron to every input, CNNs use small learnable filters that slide across the input and detect local patterns like edges, textures, and shapes. Early layers detect simple features; deeper layers combine them into complex representations like faces or objects. CNNs dramatically reduce the number of parameters compared to fully connected networks, making them efficient and highly effective for image classification, object detection, and segmentation.

Recurrent Neural Network (RNN) — `sequential data` Built for sequences where order matters — text, audio, time series, and video. Unlike feedforward networks, RNNs have loops that allow information to persist from one step to the next, giving them a form of memory. However, they struggle with long-term dependencies due to the vanishing gradient problem, where gradients shrink to near zero during backpropagation over many time steps, preventing the network from learning distant relationships.

Long Short-Term Memory (LSTM) — `solving the memory problem` An advanced type of RNN that introduces gating mechanisms — input, forget, and output gates — which control what information is stored, discarded, or passed forward. This allows LSTMs to retain relevant context over long sequences and is why they became the dominant architecture for NLP and speech recognition before transformers took over.

Transformer — `attention is all you need` The architecture behind modern AI. Rather than processing sequences step by step like RNNs, transformers process all tokens simultaneously using a mechanism called self-attention, which weighs the relevance of every word to every other word in a sequence. This makes them highly parallelizable and capable of capturing long-range dependencies with ease. Transformers are the foundation of GPT, BERT, and virtually every state-of-the-art language and vision model today.

Generative Adversarial Network (GAN) — `learning to generate` Consists of two competing networks — a generator that creates fake data and a discriminator that tries to distinguish real from fake. Through this adversarial training loop, the generator gradually learns to produce outputs indistinguishable from real data. GANs are widely used for image synthesis, style transfer, data augmentation, and deepfake generation.